

CortexCall – Data Loading & Initial Dataset Validation

1. Phase Objective

The purpose of this phase was to inspect the raw EEG dataset, verify the relationship between EEG trial files and their labels, and implement a reliable data loader that can be used by the upcoming EDA, cleaning, preprocessing, and machine-learning stages.

The main goal was to establish a trustworthy mapping:

EEG Trial → Trial ID → LEFT/RIGHT Label

2. Raw Dataset Structure

The project currently uses the following structure:

```
data/
├── labels.csv
├── MI CSV/
│   ├── cellula_MI_data_1.csv
│   ├── cellula_MI_data_2.csv
│   ├── ...
│   └── cellula_MI_data_2160.csv
```

Each EEG CSV represents one Motor Imagery (MI) trial.

The target classification problem is binary:

- left
 - right
-

3. EEG File Inspection

Before implementing the loader, the EEG filenames were inspected independently to make sure that trial numbering was reliable.

Findings

- Total EEG CSV files: **2160**
- First trial ID: **1**
- Last trial ID: **2160**
- Missing trial IDs: **None**
- Duplicate trial IDs: **None**
- Numbering starts correctly from 1.
- **Trial numbering is fully continuous.**

The filename pattern is:

cellula_MI_data_<trial_id>.csv

Examples:

cellula_MI_data_1.csv
cellula_MI_data_2.csv
cellula_MI_data_451.csv
cellula_MI_data_2160.csv

Therefore, the numeric component of each filename can safely be used as the EEG trial ID.

4. Labels File Inspection

labels.csv contains a single column representing the classification target for each trial.

The labels are:

Left , Right

The original label count is:

2400 labels

However, only:

2160 EEG signal files

are available.

Therefore:

$2400 - 2160 = 240$ extra labels

Decision Regarding Extra Labels

After discussion with the training instructor, it was confirmed that the final **240 labels can safely be discarded**.

The loader therefore keeps:

`labels[0:2160]`

and discards the remaining 240 rows.

This produces:

2160 EEG files

2160 labels

Allowing safe one-to-one alignment.

5. Data Loader Responsibilities

The `DataLoader` class is responsible for:

1. Locating `labels.csv`.
2. Locating the EEG signal directory.
3. Discovering all EEG CSV files.
4. Extracting numeric trial IDs from filenames.
5. Sorting files numerically by trial ID.
6. Loading the labels.
7. Removing the confirmed extra label rows.
8. Standardizing label formatting.
9. Aligning EEG files with their labels.
10. Building a manifest.
11. Providing a method for loading an individual EEG trial.

The loader deliberately does **not** perform EEG cleaning, filtering, feature extraction, or ML preprocessing.

Those operations belong to later stages of the pipeline.

6. Manifest

The loader constructs a manifest representing the relationship between the dataset components.

Conceptually:

trial_id	file_path	label
1	cellula_MI_data_1.c sv	left
2	cellula_MI_data_2.c sv	right
3	cellula_MI_data_3.c sv	...
...	...	...
2160	cellula_MI_data_216 0.csv	...

This manifest becomes the central reference for accessing trials during EDA and later ML stages.

7. Loader Validation Test

The loader was tested from the project environment using:

```
python -c "from data_loader import DataLoader; dl = DataLoader(data_dir='../data');  
print(len(dl)); df, label = dl.load_trial(0); print(label); print(df.shape)"
```

The resulting output confirmed:

Found 2160 signal files

Found 2400 labels and 2160 signal files

Dropping last 240 extra labels
Successfully aligned 2160 EEG trials with labels

Dataset length: 2160
First trial label: left
First trial shape: (2500, 5)

Therefore, the loader successfully performs the expected alignment and can read individual EEG trials.

8. Important Assumptions Going Forward

The current pipeline depends on several confirmed or testable assumptions:

Confirmed:

- EEG files are continuously numbered from 1 to 2160.
- There are no missing trial IDs.
- There are no duplicate trial IDs.
- Extra labels at the end of `labels.csv` can be discarded according to the project instructor.
- The loader successfully aligns 2160 signal files with 2160 retained labels.

To validate during EDA:

- Whether all trials contain exactly 2500 samples.
- Whether all trials contain the same five columns.
- The names and meanings of those columns.
- Whether EEG channels contain missing values.
- Whether values have appropriate numeric data types.
- Whether NaN, infinity, or malformed values exist.
- Whether any trials are duplicated.
- Whether LEFT and RIGHT classes are balanced.
- Whether signal ranges differ significantly between trials/channels.
- Whether extreme values or artifacts exist.
- The EEG sampling frequency and corresponding trial duration.
- Whether preprocessing such as filtering, normalization, or artifact handling is required.

9. Next Phase: Exploratory Data Analysis

The next phase should investigate the dataset at two levels.

Dataset-Level Analysis

Analyze properties across all 2160 trials, including:

- Trial dimensions
- Channel consistency
- Missing values
- Data types
- Class distribution
- Trial duration consistency
- Invalid or corrupted trials

EEG Signal-Level Analysis

Investigate the actual EEG signals, including:

- Channel statistics
- Signal distributions
- Time-domain visualization
- Signal amplitudes and ranges
- Outliers/artifacts
- LEFT vs RIGHT differences
- Potential frequency-domain analysis

The results of EDA should determine which cleaning and EEG preprocessing operations are justified rather than applying preprocessing blindly.